

7.12 (a) From conservation of charge, $\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$, and Ohm's law, $\vec{j} = \sigma \vec{E}$, we have

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = \frac{\partial \rho}{\partial t} + \sigma \nabla \cdot \vec{E} = \frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_0} \rho = 0. \text{ For a specific frequency, } \rho(\vec{x}, t) = \rho(\vec{x}, \omega) e^{-i\omega t},$$

$$\text{we can arrive at } -i\omega \rho(\vec{x}, \omega) + \frac{\sigma(\omega)}{\epsilon_0} \rho(\vec{x}, \omega) = 0, \text{ or } [\sigma(\omega) - i\omega \epsilon_0] \rho(\vec{x}, \omega) = 0$$

(b) For the equation $\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_0} \rho = 0$, the formal solution can be written as

$$\rho(t) = \int_{-\infty}^{+\infty} G(t-t') \rho(t') dt', \text{ where the Green's function is determined by}$$

$$\left(\frac{\partial}{\partial t} + \frac{\sigma}{\epsilon_0} \right) G(t) = \delta(t). \text{ Then, given an initial charge disturbance, } \rho(t) = \rho_0 \delta(t), \text{ the}$$

charge density at $t > 0$ is $\rho(t) = \rho_0 G(t)$. The Green's function can be determined by

$$G(t) = \int_{-\infty}^{+\infty} \frac{1}{-i\omega + \frac{\sigma(\omega)}{\epsilon_0}} e^{-i\omega t} d\omega.$$

$$\text{For } \sigma(\omega) = \sigma_0 / (1 - i\omega\tau), \quad G(t) = \int_{-\infty}^{+\infty} \frac{1}{-i\omega + \frac{\sigma_0}{\epsilon_0(1-i\omega\tau)}} e^{-i\omega t} d\omega = \int_{-\infty}^{+\infty} \frac{1 - i\omega\tau}{\frac{\sigma_0}{\epsilon_0} (-i\omega - \omega^2\tau)} e^{-i\omega t} d\omega$$

We can choose our integration contour in the lower half plane, for $t > 0$. The integrand has

two poles in the lower half plane, $\omega_{\pm} = -\frac{i}{2\tau} \pm \omega_p$, with $\omega_p^2 = \frac{1}{4\tau^2}$. Then,

$$\begin{aligned} G(t) &= \int_{-\infty}^{+\infty} \frac{i\omega - 1/2\tau}{(\omega - \omega_+)(\omega - \omega_-)} e^{-i\omega t} d\omega = -2\pi i \left[\frac{i\omega_+ - 1/2\tau}{\omega_+ - \omega_-} e^{-i\omega_+ t} + \frac{i\omega_- - 1/2\tau}{\omega_- - \omega_+} e^{-i\omega_- t} \right] \\ &= -2\pi i \left[\frac{-\frac{1}{2\tau} + i\omega_p}{2\omega_p} e^{-t/2\tau - i\omega_p t} - \frac{-\frac{1}{2\tau} - i\omega_p}{2\omega_p} e^{-t/2\tau + i\omega_p t} \right] \end{aligned}$$

For $\omega_p \tau \gg 1$, the Green's function becomes

$$G(t) = -2\pi i \cdot e^{-t/2\tau} \cdot \frac{i}{2} (e^{-i\omega_p t} + e^{i\omega_p t}) = 2\pi e^{-t/2\tau} \cos(\omega_p t)$$

So, the charge decays as $e^{-t/2\tau}$ and oscillates with plasma frequency ω_p .

If $\sigma(\omega) = \sigma_0$,

$$\sigma(t) = \int_{-\infty}^{+\infty} \frac{1}{-i\omega + \sigma_0/\epsilon_0} e^{-i\omega t} d\omega = \int_{-\infty}^{+\infty} \frac{i}{\omega + i\sigma_0/\epsilon_0} e^{-i\omega t} d\omega$$

$$= -2\pi i \cdot i \cdot \exp\{-\sigma_0 t/\epsilon_0\} = 2\pi \exp\{-\sigma_0 t/\epsilon_0\}$$

σ_0 , the charge disturbance just decays exponentially.